

Generation of gaseous sulfur-containing compounds in tumour tissue and suppression of gas diffusion as an antitumour treatment.

BACKGROUND AND AIMS: The mechanisms of cancer cell growth and metastasis are still not entirely understood, especially from the viewpoint of chemical reactions in tumours. Glycolytic metabolism is markedly accelerated in cancer cells, causing the accumulation of glucose (a reducing sugar) and methionine (an amino acid), which can non-enzymatically react and form carcinogenic substances. There is speculation that this reaction produces gaseous sulfur-containing compounds in tumour tissue. The aims of this study were to clarify the products in tumour and to investigate their effect on tumour proliferation. **METHODS:** Products formed in the reaction between glucose and methionine or its metabolites were analysed in vitro using gas chromatography. Flatus samples from patients with colon cancer and exhaled air samples from patients with lung cancer were analysed using near-edge x-ray fine adsorption structure spectroscopy and compared with those from healthy individuals. The tumour proliferation rates of mice into which HT29 human colon cancer cells had been implanted were compared with those of mice in which the cancer cells were surrounded by sodium hyaluronate gel to prevent diffusion of gaseous material into the healthy cells. **RESULTS:** Gaseous sulfur-containing compounds such as methanethiol and hydrogen sulfide were produced when glucose was allowed to react with methionine or its metabolites homocysteine or cysteine. Near-edge x-ray fine adsorption structure spectroscopy showed that the concentrations of sulfur-containing compounds in the samples of flatus from patients with colon cancer and in the samples of exhaled air from patients with lung cancer were significantly higher than in those from healthy individuals. Animal experiments showed that preventing the diffusion of sulfur-containing compounds had a pronounced antitumour effect. **CONCLUSIONS:** Gaseous sulfur-containing compounds are the main products in tumours and preventing the diffusion of these compounds reduces the tumour proliferation rate, which suggests the possibility of a new approach to cancer treatment.